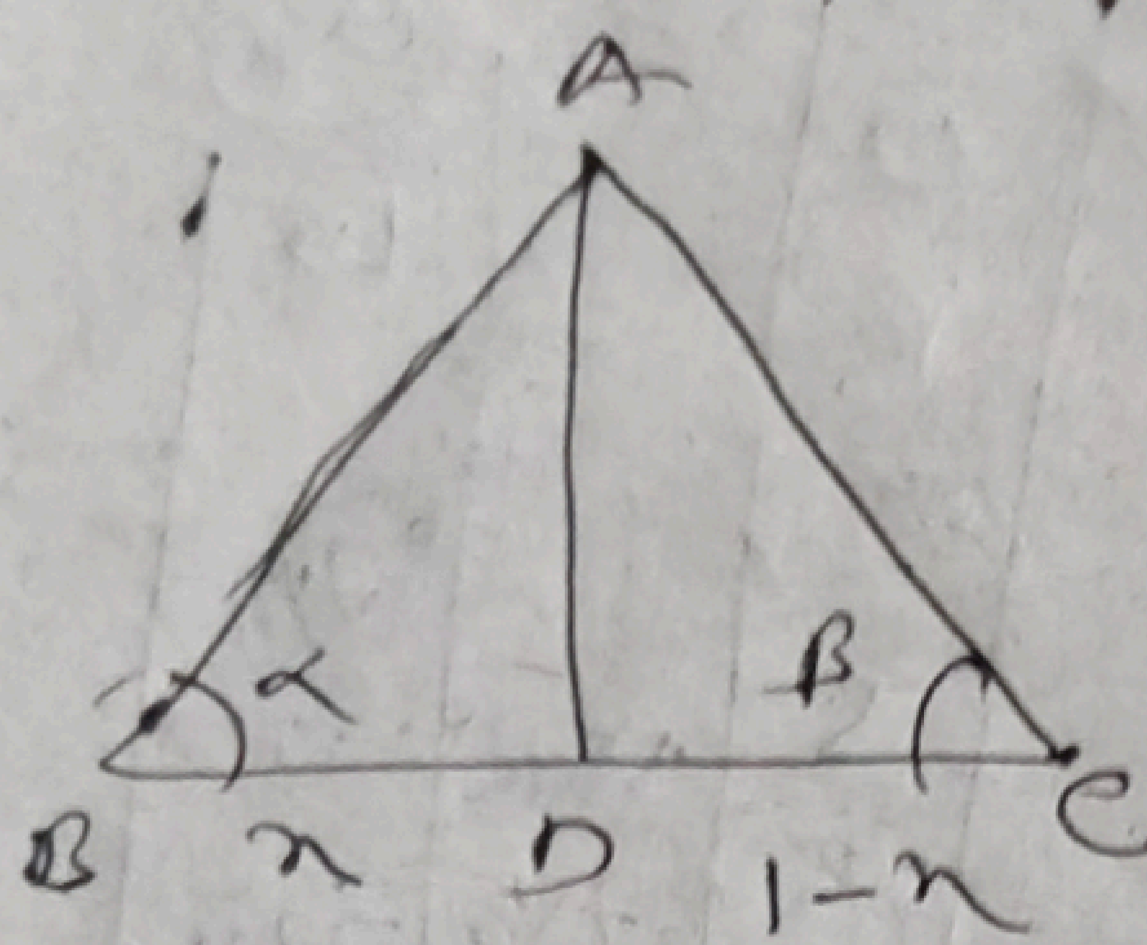

let $AD = P$ and $BD = x$
$DC = 1 - x$

$\tan\alpha = \dfrac{P}{x}$

$$\boxed{x = \frac{P}{\tan\alpha}}$$

$$\tan\beta = \frac{P}{1-x}$$

$$P = (1-x)\tan\beta$$

$$P = \tan\beta - x\tan\beta$$

$$P \Rightarrow \tan\beta - \frac{P}{\tan\alpha}\cdot\tan\beta$$

$$P + \frac{P\tan\beta}{\tan\alpha} = \tan\beta$$

$$\frac{P\tan\alpha + P\tan\beta}{\tan\alpha} = \tan\beta$$

$\tan\beta \cdot \tan\alpha = p(\tan\alpha + \tan\beta)$

$$\boxed{p = \frac{\tan\beta \cdot \tan\alpha}{\tan\alpha + \tan\beta}}$$